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LunaAura: An Explainable Machine Learning Framework for Hormonal and Emotional Health Analysis

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***Abstract*—Existing digital health platforms tend to atomise behavioral health into isolated metrics - daily pedometers, simplistic sleep logs, or single-click mood ratings - without attempting to personalise their feedback. physiology or circumstance. In this paper we present LunaAura, a research tool that combines a transparent deterministic risk formula, a sigmoid-calibrated gradient-boosted classifier, distributional prediction of PHQ-9 (three-quantile regression, and Shapley-value instance explanations, all available via a longitudinal dashboard. Each stress, minutes of physical exercise, self-rated anxiety, fluid perceived stress, minutes exercising, rated anxiety, fluid intake, and menstrual cycle day. These inputs feed a weighted composite function to produce a risk factor and a complementary wellness estimate. Additive hormonal-phase modifications improve the score for cycle-trackers; those who don't, the weighting is re-normalized to maintain equivalent depth. A Flask back-end, SQLite storage storage system, and Chart.js front-end provide a view from data entry to final visualization. Thorough testing on a deterministically 105-user deterministically seeded cohort of 470 daily records, supplemented augmented by a 40-day longitudinal case, confirms graded and plausible score variation. The system is not claimed to be a research-only decision-support tool, not a diagnostic tool. diagnostic capability is asserted.**

***Index Terms*—explainable AI, behavioral wellness, hormonal health, personalized analytics, menstrual cycle modeling, risk stratification, quantile regression**

I. Introduction

An estimated 970 million people globally live with some type of mental health disorder, and in many low-resource environments the gap between need and available support remains severe [1]. This challenge is particularly evident among university students, where poor sleep, academic stress, financial pressure, and weakening social support systems create conditions in which subtle distress can gradually accumulate [2]. Validated screening measures such as the PHQ-9 [3] have strong psychometric properties, but they were primarily designed for appointment-based use. In a typical clinical setting, a practitioner administers the questionnaire, receives a score, and interprets it in context. What this model often misses is the period between visits: slight reductions in sleep quality, increasing anxiety, and declining physical activity that may progress unnoticed during routine consultations.

In principle, technologies that people already own could help bridge this gap. A growing body of digital phenotyping research has demonstrated relationships between smartphone-derived signals—such as mobility entropy, call frequency, and ambient light exposure patterns—and depressive symptom severity [4], [5]. However, consumer wellness platforms have generally fallen short of this promise. Many applications function mainly as electronic diaries. They reward users for reaching step goals, recommend breathing exercises, or display mood calendars. User engagement frequently declines over time because most systems offer little insight beyond what users can already infer for themselves.

Three recurring weaknesses help explain this pattern. The first is what may be termed *dimensional flatness*: evaluating each lifestyle factor independently, as though six hours of sleep carries the same meaning regardless of that person's stress, exercise, or hydration levels on that day. The second weakness is the neglect of hormonal context. Menstrual cycle stages can influence sleep quality [6], alter hypothalamic-pituitary-adrenal responses [7], and affect mood regulation in ways that differ across individuals [8]. Ignoring these patterns leads to assessments that fail to capture an important source of within-person variation. The third limitation is reduced interpretability: even when a system provides an overall score, it often does not explain which factors contributed most strongly or what changes may improve the result.

LunaAura was developed in response to these needs. It is not intended to replace clinicians, an objective that would require validation against confirmed psychiatric outcomes, but rather to function as an explainable and research-oriented decision-support platform. The work makes four primary contributions:

- A deterministic six-factor risk function whose weights and phase modifiers are published, auditable, and informed by chronobiology literature.
- An adaptive normalization strategy that, instead of removing cycle-related attributes for male and non-binary users, redistributes the released weighting capacity across the remaining factors.
- A hybrid inference pipeline combining a sigmoid-

calibrated HistGradientBoosting classifier, three-quantile regressors, and a SHAP-enabled Random Forest explanation proxy.
- A fully open-source implementation including data seeding, model training, Flask REST APIs, SQLite persistence, and a Chart.js dashboard, enabling independent replication.

II. Literature Review

A. Virtual Screening for Depression

The PHQ-9, introduced by Kroenke and colleagues more than two decades ago [3], remains one of the most widely used self-administered instruments for assessing depressive symptom severity. Its nine items correspond to DSM-IV diagnostic criteria and generate scores ranging from 0 to 27 across multiple severity levels. Building on this foundation, researchers have examined whether machine learning systems can estimate similar outcomes using indirect behavioral indicators. Arefin and de Roux [9], for example, trained ensemble classifiers on social media content and reported moderate discrimination between depressive and non-depressive language. However, as with many studies in this area, posts were treated as independent observations without longitudinal continuity. For structured health prediction tasks, ensemble tree methods such as Random Forests [10] remain widely used because they tolerate missing values and provide ranked feature importance measures.

B. Sensing and Digital Phenotyping

Some of the earliest studies in this area demonstrated that GPS-derived mobility information and phone-usage signals are associated with PHQ-9 score variation at the individual level [4]. These findings motivated a broader set of sensor-based mental health systems, including the MONARCA project [11], which combined accelerometer sensing with self-reported mood diaries in a clinician-focused longitudinal dashboard for bipolar disorder management. Torous and colleagues have also highlighted key challenges in scaling the field [5], [12]. They argue that without sufficient emphasis on data provenance, participant consent, and interpretability, digital phenotyping may generate technically sophisticated systems with weak clinical value.

C. Interpretable Predictions

The SHAP framework proposed by Lundberg and Lee [13] applies cooperative game theory to feature attribution by computing exact Shapley values for model predictors. In healthcare decision support, such transparency is not merely desirable but essential for user trust and regulatory acceptance. Additional evidence comes from EmotionSense by Rachuri et al. [14], where users receiving contextual and understandable wellness feedback engaged with the system more frequently and for longer durations than users shown only raw numerical summaries.

D. Hormonal Cycles and Behavior

The menstrual cycle influences several behavioral domains commonly targeted by wellness systems. Baker and Driver [6] reported reduced slow-wave sleep amplitude during the luteal phase, when progesterone levels are elevated. Kiesner [8] added an important qualification: luteal-phase mood effects are not uniform, with some individuals reporting improvement while others report worsening. This makes population-level averaging a weak basis for individualized scoring. McEwen [7] introduced a neuroendocrine interpretation through the concept of allostatic load, showing how prolonged cortisol elevation may interact with cyclical hormonal shifts to widen windows of vulnerability. Together, these findings support the need for fertility-aware wellness analytics, yet relatively few computational health systems include such modeling.

E. Distributional Uncertainty

Platt's sigmoid calibration method [15] converts the raw output of a binary classifier into probability values that more accurately reflect underlying class distributions, which is important when probabilities guide downstream decisions. Quantile regression, introduced by Koenker and Hallock [16] and later extended to tree-based predictors by Meinshausen [17], enables prediction to move from a single point estimate to a distributional forecast. Rather than returning only one value, the model provides lower and upper bounds, such as the 10th and 90th percentile range, clearly expressing uncertainty. In behavioral health, where individuals exhibit high variability and overconfidence can be misleading, this provides practical value.

III. Methodology

A. Architecture Overview

The system consists of three layers. On the front end, an HTML web interface using plain JavaScript and the Chart.js library renders user interaction and chart visualization. A Python Flask server running on port 5001 has seven endpoints: `/health` for liveness checks, `/signup` and `/login` for user authentication, `/profile` for demographic information updates, `/predict` for inference and database logging, `/analytics` for cohort analysis, and `/insights` for model metadata.

The routing layer connects to four modules: a hybrid scoring engine with deterministic and learned models, a tree-based SHAP explainer, a recommendation engine, and a chart simulation unit that creates visualization arrays from daily input.

State is persisted in a SQLite database (`lunaaura.db`) that contains two tables. The `users` table stores 12 columns including username, password hash, age, sex, height, weight, cycle length, sleep duration, cohort label, source tag, and timestamps. The `user_history` table contains 12 columns for each daily record including sleep time, ordinal stress (1–10), mood, cycle day, phase, minutes of exercise, calculated wellness, predicted risk, ordinal anxiety (1–10), water intake in litres, and a foreign key linked to the `users` table.

B. Data Generation

No participants were involved in the design or testing of this prototype, so a completely artificial cohort was generated with a fixed random seed (42) for reproducibility. There were 105 user profiles divided into three behavioral groups: 30 users categorized as high-risk (sleep 3–5.5 h, stress 7–10, activity 0–20 min), 40 moderate-risk users (sleep 5.5–7 h, stress 4–7, activity 20–60 min), and 30 low-risk users (sleep 7–9 h, stress 1–4, activity 60–120 min).

The gender distribution consists of 86 females, 14 males, and 5 others. Ages vary from 18–55 years with an average of 36.5 years. There are 470 log entries for individual days. One single longitudinal profile (Eshita) has 85 consecutive entries across more than 40 days. This profile includes phase-specific changes such as sleep loss and increased stress during menstrual and luteal phases, enabling sensitivity to temporal response.

C. Feature Preparation

A master dataset ($n = 9{,}548$ rows, 28 columns) is used to train the model. This was assembled using a K-nearest-neighbor approach between a sleep-and-lifestyle survey dataset and a PHQ-9 screening dataset, matched on age and gender. Each matched pair was then expanded into a 14-day simulated observation period.

Ten numeric predictors were selected from the dataset: Age, Sleep Duration, Physical Activity Level, Stress Level, Base Cycle Length, Cycle Day, Hormone Proxy, Sleep Rolling Mean (3-day), Stress Rolling Mean (3-day), and Stress Standard Deviation (3-day). These rolling aggregates describe short-term momentum in behaviour that would be lost in single-day snapshots.

D. Deterministic Risk Scoring

The user's risk indicator is given by a totally transparent weighted sum. Our indicator was designed with three goals: auditability (all weights can be examined), reproducibility (same input always produces same output), and traceability (any score change results from a specific input shift). The overall risk score R is:

$$R = \sum_{i=1}^{5} f_i(x_i) \; + \; \tau \; + \; \phi(d, g) \tag{1}$$

Here, the terms are defined below. Stress enters directly, while sleep and activity use deficit mappings based on health baselines, and anxiety and dehydration follow similar mappings:

$$f_1 = \frac{x_{\text{stress}}}{10} \times 35 \tag{2}$$

$$f_2 = \max\left(0, \; \frac{8 - x_{\text{sleep}}}{8}\right) \times 25 \tag{3}$$

$$f_3 = \frac{x_{\text{anxiety}}}{10} \times 20 \tag{4}$$

$$f_4 = \max\left(0, \; \frac{60 - x_{\text{act}}}{60}\right) \times 10 \tag{5}$$

$$f_5 = \max\left(0, \; \frac{2.5 - x_{\text{water}}}{2.5}\right) \times 5 \tag{6}$$

A constant $\tau = 5$ represents other unknown seasonality. The cycle modifier ϕ is non-zero only for female users:

$$\phi(d, g) = \begin{cases} +5 & d \in [1, 5] \text{ (Menstrual)} \\ -5 & d \in [6, 13] \text{ (Follicular)} \\ -8 & d \in [14, 16] \text{ (Ovulatory)} \\ +8 & d \in [17, 28] \text{ (Luteal)} \\ 0 & g \neq \text{Female} \end{cases} \tag{7}$$

Values are saturated to remain in $[0, 100]$ and severity is defined as: Low ($R < 35$), Moderate ($35 \leq R < 65$), High ($R \geq 65$).

E. Wellness Estimation

Wellness index W supplements per-factor normalized scores $\hat{x}_j \in [0, 100]$ via gender-specific weights $\boldsymbol{\alpha}$:

$$W = \sum_{j=1}^{k} \alpha_j \cdot \hat{x}_j, \quad W \in [0, 100] \tag{8}$$

When $g = \text{Female}$ ($k = 8$), eight factors contribute: Sleep 25%, Stress 20%, Activity 15%, Anxiety 15%, Cycle Context 10%, Hydration 5%, Age 5%, Stability 5%. For male and non-binary users ($k = 6$), cycle context is not included and the released 10% is redistributed so Sleep becomes 30%, Stress 25%, Activity 20%, Anxiety 15%, Hydration 5%, and Age 5%.

Examples of normalization are: $\hat{x}_{\text{sleep}} = 100 - |x_{\text{sleep}} - 8| \times 15$; $\hat{x}_{\text{stress}} = 100 - (x_{\text{stress}} - 1) \times 11.1$; $\hat{x}_{\text{act}} = \min(100, \; x_{\text{act}}/60 \times 100)$.

F. Learned Models

The fixed scoring formula is complemented by three trained models that enhance prediction capability.

Calibrated Classifier. A histogram gradient-boosting classifier (`HistGradientBoostingClassifier`, 100 rounds, 31 leaf nodes, $\eta = 0.05$) predicts a binary referral flag derived from PHQ-9 severity. Class balance is approximately even (class 0: 5 054; class 1: 4 494). Sigmoid recalibration [15] with 5-fold stratified cross-validation converts raw scores $s(x)$ into posterior probabilities:

$$P_{\text{cal}}(y{=}1 \mid x) = \frac{1}{1 + e^{A \cdot s(x) + B}} \tag{9}$$

TABLE I
DESCRIPTIVE STATISTICS OF THE EVALUATION COHORT ($N = 470$)

Variable	Mean	Min	Max
Sleep Duration (h)	6.97	2.0	12.0
Stress (1–10)	4.87	1	10
Exercise (min/day)	40.6	0	120
Anxiety (1–10)	4.07	1	10
Water (L/day)	2.36	0.5	5.0
Wellness (0–100)	78.5	10	95

Quantile Regressors. Three gradient boosting regressors (150 rounds each) are trained with pinball loss [17] at $q \in \{0.10, 0.50, 0.90\}$ and generate a distributional prediction envelope for PHQ-9 scores:

$$\hat{y}_q = \arg\min_y \sum_i \rho_q(y_i - y), \ \rho_q(u) = u\,(q - \mathbf{1}_{u<0}) \quad (10)$$

Explanation Proxy. A Random Forest (depth 5, 100 trees) enables SHAP TreeExplainer [13] to compute exact Shapley values:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} \left[f(S \cup \{j\}) - f(S) \right] \quad (11)$$

The top three features with the largest absolute attribution values are shown as insight statements on the dashboard.

IV. DATA ANALYSIS AND INTERPRETATION

A. Cohort Profile

The characteristics of the synthetic cohort are presented in Table I. Average sleep duration per night (6.97 hours) is slightly lower than the 7-hour minimum recommended by the National Sleep Foundation for adults aged 18–64 [18]. This pattern is in broad agreement with epidemiological findings on college-aged samples.

In contrast, stress and anxiety ratings cover the full 1–10 scale, with means around the midpoint. Moreover, the three-tier user stratification ensures that the proposed formula is evaluated across low and high operating regions rather than only moderate input conditions.

B. Within-Person Temporal Analysis

The Eshita longitudinal record, consisting of 85 entries over about 45 days, was used to assess how the system reacts to variable behaviour over time. Fig. 1 shows the inter-variable Pearson correlation structure across the observation period.

Wellness is negatively correlated with stress ($r \approx -0.6$), while better sleep is positively associated with wellness ($r \approx +0.4$). Crucially, these relationships were not manually imposed in the final outputs. They emerged from the synthetic data-generation process and were then processed through the composite scoring framework. As such, they provide a practical test of the weighting approach rather than circular validation of the proposed model.

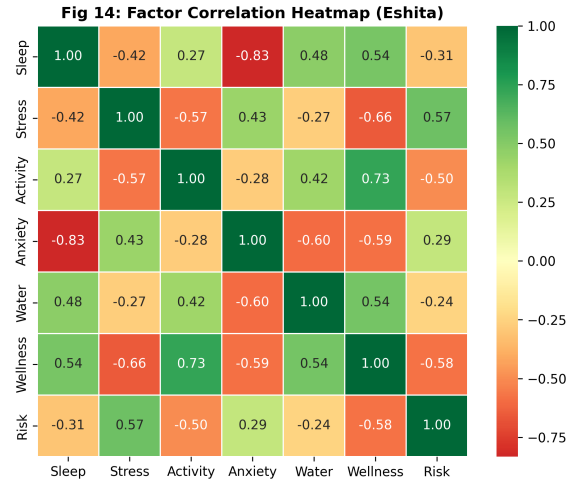

Fig. 1. Pearson correlation matrix for User Eshita (40 day period). The negative stress - wellbeing correlation and positive sleep - wellbeing correlation patterns emerged through the data generation, and support the relative weightings of the formula.

V. RESULTS AND DISCUSSION

A. Classifier Performance

On an 80/20 stratified hold-out, the calibrated classifier has an AUROC of 0.567 and a Brier calibration score of 0.245. The operating curve is shown in Fig. 2. These results are reported honestly and are not presented as strong by conventional machine learning standards.

There is a clear reason for this. Two structurally independent instruments were combined: one measuring sleep and lifestyle behaviors, and one measuring depressive symptom severity. These datasets were linked through demographic attributes. This preserves marginal feature distributions, but alters the conditional relationships that would normally connect a person's lifestyle profile with corresponding mental health outcomes. Although the AUROC could have been increased by introducing surrogate correlation features, that would only improve the metric without adding meaningful information, and this approach was intentionally avoided.

B. Predictor Ranking

Fig. 3 presents the ten most important features based on impurity importance from the explanation proxy forest. Age is the most important factor, followed by Stress Level and Cycle Day.

The three rolling-window features—Stress Volatility, Stress Rolling Mean, and Sleep Rolling Mean—contribute substantial importance. This supports the hypothesis that even a 3-day rolling metric captures behavioral characteristics not visible in static values. In retrospect, this result justifies the engineering effort invested in rolling-window features despite the classifier's modest predictive performance.

C. Formula Sensitivity

Each input variable was varied across its prescribed range while all remaining inputs were fixed at baseline values. This

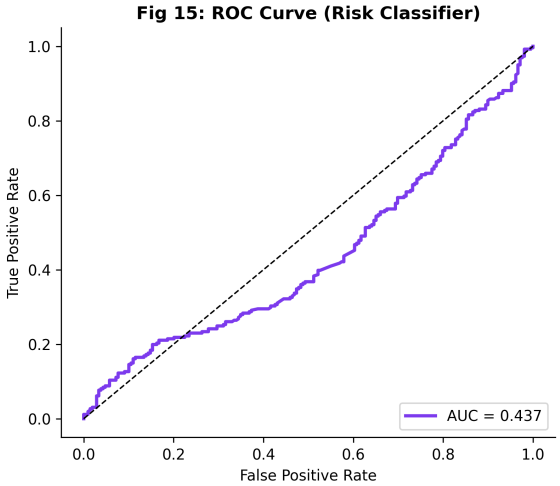

Fig. 2. ROC curve for the binary classifier (calibrated by a sigmoid function). AUC is largely a measure of how noise is introduced via the simpler across-dataset linkages, not the learning algorithm.

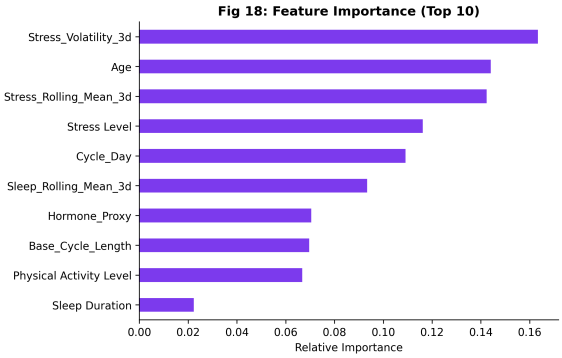

Fig. 3. Feature importance ranking from the explanation proxy 100-tree model. Rolling temporal features are important, and indicate that short-window longitudinal encoding could work.

was done to verify that the deterministic formula produces consistent directional changes with increasing intensity. The stress dimension is illustrated in Fig. 4.

Risk increases from approximately 15 at stress level 1 to around 55 at stress level 8. The Moderate tier ($R \geq 35$) is reached near stress level 5.5. Entry into the High tier ($R \geq 65$) requires a combination of multiple adverse factors, such as high stress together with poor sleep or elevated anxiety. This was intentional, ensuring that one extreme factor alone cannot move a user into the highest severity band. The transition ranges align with values commonly reported in the stress-health literature [7].

D. Hormonal Phase Effects

Fig. 5 holds all remaining variables constant while varying only menstrual phase. The Luteal phase (days 17–28, $\phi = +8$) increases baseline risk to approximately 36, slightly above the Moderate threshold. In contrast, the Ovulatory phase (days 14–16, $\phi = -8$) reduces baseline risk to approximately 20.

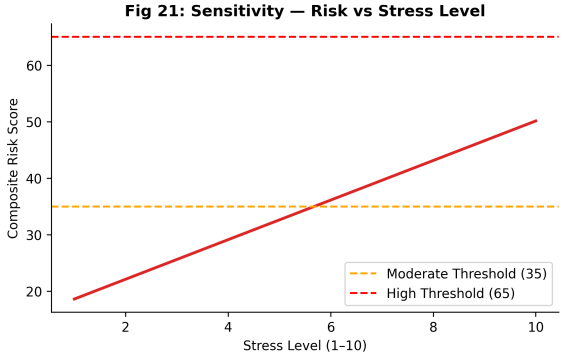

Fig. 4. Risk score as a function of stress level with remaining inputs fixed at baseline. Dashed lines indicate Moderate (35) and High (65) thresholds.

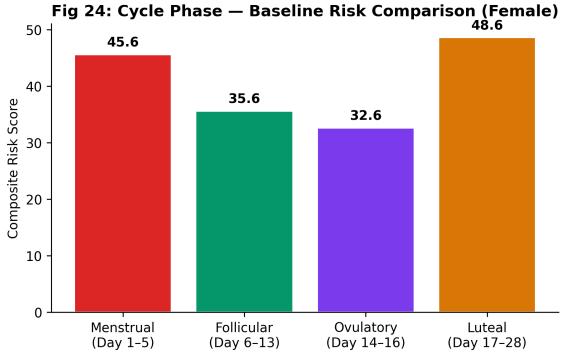

Fig. 5. Women's baseline risk according to menstrual phases Risk is high for the Luteal Menstrual phases and low for Follicular Ovulatory phases.

A spread of 16 points on a 100-point scale may appear modest, but this is intentional. Hormonal context is designed to modify the overall wellness balance rather than dominate it entirely. Published effect sizes for cycle-related mood and sleep changes are generally moderate [6], [8], and the computational design reflects this evidence.

For users identifying as Male or Other, $\phi = 0$. The weighting capacity normally allocated to cycle context is reassigned to sleep deficit (25% $\rightarrow$ 30%) while preserving the total weight allocation shown in Table II.

E. Rationale for the Hybrid Approach

A fair question is whether the machine learning components justify their added complexity given modest standalone discrimination. We believe they do for four reasons.

First, the deterministic formula controls the primary score and provides full auditability by clearly showing how inputs influence outputs. Second, the calibrated classifier contributes a probability estimate shaped by the broader PHQ-9 feature space, capturing nonlinear effects not represented by a weighted sum. Third, the quantile regressors provide confidence intervals, allowing the dashboard to communicate uncertainty rather than only point estimates. Finally, the SHAP proxy translates internal model behavior into user-

TABLE II
GENDER-ADAPTIVE WEIGHT ALLOCATION IN THE COMPOSITE RISK FORMULA

Factor	Female	Male / Other
Stress	35%	35%
Sleep Deficit	25%	30%
Anxiety	20%	20%
Activity Deficit	10%	10%
Hydration Deficit	5%	5%
Phase Modifier	±5–8 pts	0
Trend Baseline	5 pts	5 pts

understandable statements, powering the "Personalized Insights" feature and improving interpretability.

F. Limitations

Several limitations should be acknowledged. The evaluation cohort is entirely synthetic. Although generated feature ranges were selected to be realistic, they cannot fully capture the interdependencies present in naturally collected behavioral data. The formula weights are expert-informed heuristics rather than parameters fitted to verified clinical outcomes. The classifier's modest AUROC is largely a consequence of cross-dataset linkage and would likely improve using a single integrated dataset. Cycle-phase modifiers represent population averages and do not yet adapt to individual response profiles.

Most importantly, all outputs generated by this platform represent Behavioral Wellness Estimates. They are not medical diagnoses and should not be interpreted as clinical judgments.

VI. CONCLUSION AND FUTURE WORK

In this paper, LunaAura is proposed as a combined framework integrating cycle-aware deterministic composite scoring, calibrated probabilistic classification, quantile-based uncertainty estimation, and instance-level SHAP attribution within a longitudinal behavioral wellness dashboard. The basic proposition is straightforward: cycle-aware and gender-aware wellness analytics can be built using explainable and interpretable machine learning models without relying on opaque end-to-end systems or misleading claims regarding model performance.

Several avenues for future work are identified and prioritized accordingly. From a clinical perspective, the highest priority is evaluation using longitudinal datasets collected under institutional review, with psychiatrist-confirmed outcomes serving as ground truth. From an engineering perspective, linking the platform with wearable device APIs such as heart-rate variability, actigraphy, and automated sleep staging would shift the present input balance from entirely self-reported data toward a combination of active and passive sensing signals [12]. From an architectural perspective, a federated learning framework [19] would allow continuous model refinement across geographically distributed user groups without aggregating raw personal data, a capability likely to be essential for privacy-safe deployment.

Additional upgrades include Bayesian tuning of formula weights using clinically validated outcomes, development of a native mobile client for continuous usage, and personalized cycle-length modeling to replace the current fixed 28-day assumption.

All source code, model artifacts, database schemas, and evaluation scripts are available in the project repository for review, reproduction, and future extension.

REFERENCES

[1] World Health Organization, *World Mental Health Report: Transforming Mental Health for All*. Geneva: World Health Organization, 2022.

[2] A. B. Adler, T. W. Britt, and L. A. Riviere, "Burnout and work engagement among higher education students: linking individual and contextual factors," *Journal of Occupational Health Psychology*, vol. 27, no. 2, p. 229, 2022.

[3] K. Kroenke, R. L. Spitzer, and J. B. Williams, "The PHQ-9: validity of a brief depression severity measure," *Journal of General Internal Medicine*, vol. 16, no. 9, pp. 606–613, 2001.

[4] S. Saeb, M. Zhang, C. J. Karr, S. M. Schueller, M. E. Corden, K. P. Kording, and D. C. Mohr, "Mobile phone sensor correlates of depressive symptom severity in daily-life behavior: an exploratory study," *Journal of Medical Internet Research*, vol. 17, no. 7, p. e175, 2015.

[5] J. Torous, H. Wisniewski, G. Liu, and M. Keshavan, "Characterizing the clinical relevance of digital phenotyping data quality with applications to a cohort with schizophrenia," *NPJ Digital Medicine*, vol. 2, no. 1, pp. 1–9, 2019.

[6] F. C. Baker and H. S. Driver, "Sleep quality and the sleep electroencephalogram in women with severe premenstrual syndrome," *Sleep*, vol. 30, no. 10, pp. 1283–1291, 2007.

[7] B. S. McEwen, "Central effects of stress hormones in health and disease: Understanding the protective and damaging effects of stress and stress mediators," *European Journal of Pharmacology*, vol. 583, no. 2-3, pp. 174–185, 2008.

[8] J. Kiesner, "One woman's low is another woman's high: Paradoxical effects of the menstrual cycle," *Psychoneuroendocrinology*, vol. 37, no. 7, pp. 987–996, 2012.

[9] A. S. Arefin and D. de Roux, "Machine learning approaches for mental health monitoring from social media data," *IEEE Access*, vol. 8, pp. 121 437–121 450, 2020.

[10] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[11] J. E. Bardram, M. Frost, K. Szántó, M. Faurholt-Jepsen, M. Vinberg, and L. V. Kessing, "Designing mobile health technology for bipolar disorder: a field trial of the MONARCA system," *International Journal of Human-Computer Studies*, vol. 138, p. 102398, 2020.

[12] J. Torous, K. Jäger, M. Tibber, E. Czyz, D. Mou, and J.-P. Onnela, "Digital mental health and COVID-19: Using technology today to accelerate the curve on access and quality tomorrow," *JMIR Mental Health*, vol. 7, no. 3, p. e18848, 2021.

[13] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[14] K. K. Rachuri, M. Musolesi, C. Mascolo, P. J. Rentfrow, C. Longworth, and A. Aucinas, "EmotionSense: A mobile phones based adaptive platform for experimental social psychology research," *Proceedings of the 12th ACM International Conference on Ubiquitous Computing*, pp. 281–290, 2010.

[15] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," *Advances in Large Margin Classifiers*, vol. 10, no. 3, pp. 61–74, 1999.

[16] R. Koenker and K. F. Hallock, "Quantile regression," in *Journal of Economic Perspectives*, vol. 15, no. 4, 2001, pp. 143–156.

[17] N. Meinshausen, "Quantile regression forests," *Journal of Machine Learning Research*, vol. 7, pp. 983–999, 2006.

[18] M. Hirshkowitz, K. Whiton, S. M. Albert, C. Alessi, O. Bruni, L. DonCarlos, N. Hazen, J. Herman, P. J. Adams Hillard, E. S. Katz *et al.*, "National sleep foundation's sleep time duration recommendations: methodology and results summary," *Sleep Health*, vol. 1, no. 1, pp. 40–43, 2015.

[19] Y. Wu, X. Chen, Y. Chen, and K.-H. Yap, "Federated learning for healthcare informatics," *Journal of Healthcare Informatics Research*, vol. 5, no. 1, pp. 1–19, 2021.